

industry, let us try to understand the cloud from a technical perspective. A generalised architecture of a cloud can be conceptualised as consisting of two major components — the front-end and the back-end.

The front-end contains the client infrastructure, i.e., the device and the user interface of the application used for communicating with the cloud. In a real-world context, your smartphone and the Google Drive app are the front-end client infrastructure that can be used for accessing the Google cloud.

The back-end contains the cloud infrastructure, i.e., all the mechanisms and machinery required to run a cloud computing service. The servers, virtual machines, services and storage are all provided by the cloud infrastructure, as shown in Figure 1. Let's quickly go through each component of the back-end in order to get a complete picture.

- **Application:** The back-end of whatever app the user or business uses to interact with the cloud via the Internet.
- **Service:** The infrastructure for the type of service that the cloud provides. We are going to go into detail about all the different types of services in this article.
- **Runtime:** The provision of runtime and execution made available to the virtual machines.
- **Storage:** The acquisition and management of user/business data with the flexibility of scaling.
- **Infrastructure:** The hardware and software required to run the cloud.
- **Security and management:** Putting security mechanisms in place to protect user/business data as well as managing individual units of the cloud architecture to avoid overload and service outages.

Software as a Service (SaaS)

Software as a Service is a cloud computing model that provides software and applications as a service

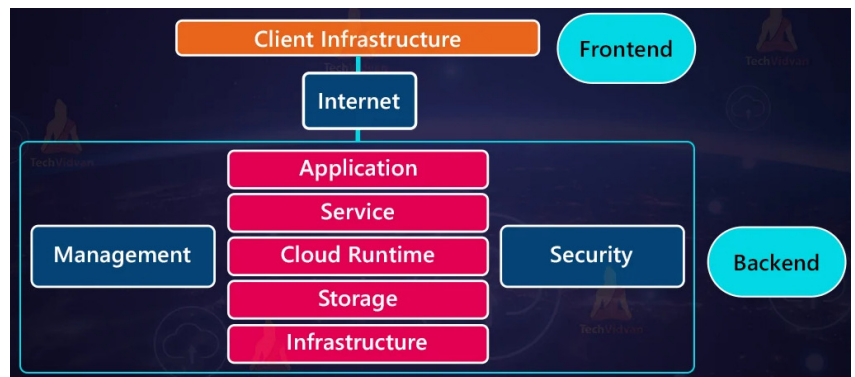


Figure 1: The architecture of a cloud (Courtesy: TechVidvan)

over the Internet. A good example of this is Google Drive or Google Workspace. All the apps available in Google Drive such as docs, sheets, slides, forms, etc. can be accessed online using a Web browser and saved automatically to the cloud. You can access the latest version of your documents through any device. All you need to do is login to your account. This is the benefit of having the Software as a Service model. Instead of having to install anything to your device locally or using your local storage space, you can directly access the software application in the cloud thus removing a lot of the liabilities that come with localised software. SaaS often follows the 'pay as you go' model, i.e., you pay for the services you need. You can always purchase more storage and/or features by paying more or downgrade your package as per your requirements.

Benefits of SaaS

1. SaaS is highly scalable, thanks to the 'pay as you go' model. You can increase/decrease storage and/or the features of the apps as and how you need to.
2. It is considerably cost-effective given the features it provides such as real-time access through any device with any operating system.
3. It involves low effort at the customer-end. No installations or confusing steps are required to

initialise the software. You can use it from the comfort of your browser and/or app.

4. Software updates automatically without you having to install it or wait for installation at your end.

Platform as a Service (PaaS)

Not every tech startup has the required resources to maintain their own infrastructure to run their apps on the cloud. In many cases, companies (especially startups) prefer to have their app hosted on the cloud without having to handle all the backend infrastructure. It is in situations such as these where a Platform as a Service model comes into play. Companies such as Heroku cloud offer PaaS architecture-based cloud solutions for companies and individuals to host and run their apps in the cloud without having any direct contact with the hardware infrastructure. Like SaaS, this model also provides flexibility in choosing only the services you require along with scalability and security from an infrastructural perspective.

Benefits of PaaS

1. No hassle of handling the cloud infrastructure. You outsource that to the company that hosts your app in their cloud. This helps you focus solely on your app development life cycle.
2. PaaS is scalable. You can increase or decrease your storage requirements, add-on services, etc,